

# Clinical Outcomes After Supraorbital or Infraorbital Transcutaneous Electrical Nerve Stimulation for Migraine Prevention: An Exploratory Three-Arm Randomized, Sham-Controlled Trial

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Mandatory pre-submission verification: Replace all bracketed fields. Verify the randomization method against the original allocation file: a previous draft reported fixed blocks of four for a 1:1:1 three-arm allocation, which is mathematically incompatible with balanced permuted-block randomization. Insert the correct method only after source verification. Ethics approval and trial-registration identifiers are also indispensable.

## Abstract

**Background:** External trigeminal nerve stimulation is a non-invasive option for migraine prevention. Evidence has primarily concerned supraorbital stimulation; randomized evidence for infraorbital stimulation and direct comparisons between stimulation sites is limited.

**Objective:** To estimate clinical outcomes after supraorbital or infraorbital transcutaneous electrical nerve stimulation (TENS), compared with inert sham stimulation, in adults with migraine. The comparison between active sites was exploratory.

**Methods:** In this single-center, three-arm randomized trial, 83 adults meeting International Classification of Headache Disorders, third edition, criteria for migraine were allocated to bilateral supraorbital TENS ( $n = 28$ ), bilateral infraorbital TENS ( $n = 27$ ), or sham stimulation ( $n = 28$ ). Active TENS was administered in clinic at 100 Hz and 200  $\mu$ s, at a strong but tolerable sensory intensity (target approximately 10 mA), for 30 minutes per session, six sessions per week for four weeks. Sham electrodes were placed supraorbitally but delivered no current. The primary outcome was the number of migraine attacks during the month assessed at three months. Secondary outcomes included pain intensity, Migraine Disability Assessment (MIDAS), Headache Impact Test-6 (HIT-6), acute oral analgesic-use frequency, and Migraine Prevention Questionnaire score. Analyses available from the supplied study report were unadjusted and should be considered exploratory.

**Results:** All 83 randomized participants provided three-month data. Mean (SD) monthly attack frequency changed from 14.18 (3.95) to 6.59 (2.70) in the supraorbital group, from 15.13 (4.51) to 6.16 (2.48) in the infraorbital group, and from 12.32 (4.99) to 12.94 (5.01) in the sham group. At three months, unadjusted mean differences versus sham were 6.35 attacks/month (approximate 95% CI, 8.46 to 4.24) for supraorbital TENS and 6.78 (8.86 to 4.70) for infraorbital TENS. The unadjusted difference between active groups was 0.43 (0.94 to 1.80). Secondary outcomes showed the same descriptive pattern. No serious adverse events were reported; transient itching and paresthesia were common in the active groups.

**Conclusions:** In this selected cohort able to complete an intensive in-clinic schedule, outcomes at three months favored both active TENS protocols over an inert supraorbital sham. The data do not establish equivalence of the two active sites. Interpretation is limited by unresolved randomization-reporting details, absence of baseline-adjusted longitudinal analyses, an inactive and site-mismatched sham, uncertain participant masking, selection before randomization, and short follow-up. Confirmatory, adequately powered trials with credible site-matched sham stimulation and prespecified analyses are required.

**Trial registration:** [Registry, identifier, date of registration; verify that registration preceded enrollment]

**Ethics approval:** [Committee name and approval identifier]

**Keywords:** migraine; external trigeminal nerve stimulation; transcutaneous electrical nerve stimulation; supraorbital nerve; infraorbital nerve; randomized trial

## 1. Introduction

Migraine is a disabling neurological disorder and a major contributor to years lived with disability worldwide [1,2]. Recurrent attacks may include moderate-to-severe headache, nausea, photophobia, and phonophobia and can substantially impair occupational, social, and family functioning [3]. Preventive pharmacotherapy is effective for many patients, but incomplete response, adverse effects, contraindications, and poor persistence create a need for additional treatment options [4].

Non-invasive neuromodulation seeks to modify nociceptive processing without systemic drug exposure. External trigeminal nerve stimulation, delivered through cutaneous electrodes, has been studied for acute and preventive migraine treatment [5–8]. Most evidence concerns stimulation of the supraorbital and supratrochlear branches of the ophthalmic division of the trigeminal nerve. A sham-controlled trial of a supraorbital device reported preventive benefit, and subsequent studies and reviews have generally suggested favorable tolerability, although effect

estimates and certainty vary [5–10].

The infraorbital nerve, a terminal branch of the maxillary division, provides another accessible trigeminal target. Evidence for infraorbital stimulation in migraine is sparse and has largely arisen from nerve-block or invasive stimulation studies rather than randomized trials of surface TENS [11,12]. Whether surface stimulation at this site produces clinically relevant benefit, and whether outcomes differ from supraorbital stimulation, remains uncertain.

This three-arm trial estimated outcomes after supraorbital TENS, infraorbital TENS, or inert sham stimulation. The primary objective was to compare each active intervention with sham for monthly migraine attack frequency at three months. The direct comparison between active sites was exploratory; the trial was not designed as an equivalence or non-inferiority study.

## **2. Methods**

### **2.1 Study design and setting**

This was a single-center, parallel-group, three-arm randomized, sham-controlled trial conducted in the outpatient Physical Medicine and Rehabilitation clinics of Isfahan University of Medical Sciences, Iran, between [insert exact first and last enrollment dates]. The planned allocation ratio was 1:1:1. Reporting follows CONSORT guidance for randomized and non-pharmacological trials [13,14].

The institutional ethics committee approved the protocol ([insert committee and approval identifier]). The trial was registered at [insert registry, identifier, and registration date]. These identifiers and prospective registration status must be verified against source documents before submission. All participants provided written informed consent.

### **2.2 Participants**

Adults aged 18–65 years were eligible if they met ICHD-3 criteria for migraine with or without aura and reported at least two migraine attacks per month during the preceding three months [3]. A migraine attack was defined as a qualifying migraine episode; a subsequent episode was counted as a new attack after a pain-free interval of at least 24 hours. Participants recorded attacks in a prospective headache diary. The report should specify the length of the baseline diary period and the exact 28- or 30-day windows used at follow-up.

Exclusion criteria were preventive migraine medication within two weeks before enrollment; nerve block, botulinum toxin, or another preventive intervention within the preceding year; medication-overuse headache; another concurrent headache disorder; severe psychiatric or systemic illness; pregnancy or lactation; substance misuse; or anticipated inability to complete the intervention schedule. The two-week medication washout may have been insufficient for drugs with long half-lives and is therefore considered a limitation.

Of 137 screened individuals, 54 were excluded before randomization because they were unable or unwilling to attend six clinic sessions per week for four weeks. The resulting sample was selected for schedule flexibility and anticipated adherence and should not be regarded as representative of all patients seeking migraine prevention.

### **2.3 Randomization and allocation concealment**

The supplied report states that a computer-generated sequence allocated participants in a 1:1:1 ratio and that assignments were placed in sequentially numbered, opaque, sealed envelopes. The original sequence-generation record must be checked before submission. The prior statement that fixed blocks of four were used cannot produce balanced 1:1:1 allocation across three groups. The final manuscript must report the actual restriction method (simple randomization or valid permuted blocks), block sizes if applicable, who generated the sequence, who prepared the envelopes, who enrolled participants, and when envelopes were opened.

### **2.4 Masking**

Participants were informed that they would receive one of three stimulation protocols. Active treatment produced paresthesia, whereas sham delivered no current; participant masking was therefore unlikely to be fully credible. Treating therapists could not be masked. Outcome assessment and statistical analysis were reported as masked to allocation, but the manuscript should identify these personnel and describe how masking was maintained.

After treatment, participants guessed their assignment. Correct guesses were reported for 62% of active-treatment participants and 54% of sham participants ( $p = 0.34$ ). A non-significant difference does not demonstrate successful masking; these data are reported descriptively.

### **2.5 Interventions**

**Supraorbital TENS:** Surface electrodes were placed bilaterally over the supraorbital foramina/notches. A Novin Co. (Iran) TENS device delivered 100-Hz stimulation with a pulse width of 200  $\mu$ s for 30 minutes. Current was increased to a strong but tolerable paresthesia, targeting approximately 10 mA according to individual tolerance.

**Infraorbital TENS:** Surface electrodes were placed bilaterally over the infraorbital foramina. Device, frequency, pulse width, session duration, intensity target, and treatment schedule were identical to those used in the supraorbital

group.

Sham stimulation: Electrodes were placed over the supraorbital nerves for the same session duration and schedule, but no current was delivered. Thus, sham was inert rather than sensory-matched and matched the electrode site only for the supraorbital comparison.

All protocols comprised 24 supervised sessions (six per week for four weeks). Participants could use acute medications, including acetaminophen, non-steroidal anti-inflammatory drugs, or ergot derivatives, as clinically required. The protocol did not standardize rescue medication, and changes in medication type or dose were not reported.

## **2.6 Outcomes**

Outcomes were recorded at baseline, one month, and three months.

The primary outcome was monthly migraine attack frequency at three months. Contemporary International Headache Society guidance generally recommends change in monthly migraine days, moderate/severe headache days, or a prespecified responder rate for preventive trials [15]. Attack frequency is clinically relevant but limits comparability with current trials. The manuscript should report migraine-day data if they were collected.

The secondary outcomes were:

1. pain intensity on a 0–10 visual analogue scale (VAS);
2. migraine-related disability measured with MIDAS;
3. headache impact measured with HIT-6;
4. number of occasions of oral acute-analgesic use per month; and
5. the reported Migraine Prevention Questionnaire (MPQ) score.

The exact VAS recall period and scoring rules must be stated. MIDAS ordinarily refers to disability over the preceding three months; interpreting a one-month administration requires justification because recall windows overlap. The exact MPQ version, item wording, scoring range, direction of benefit, language validation, and citation must also be verified. The values reported in the supplied manuscript cannot be interpreted confidently without this information.

Adverse events were elicited through open-ended questioning at visits. No standardized checklist, severity grading, causality assessment, or predefined safety window was used.

## **2.7 Statistical analysis**

The supplied report summarized continuous variables using means and SDs, tested within-group change using Friedman tests, and compared groups at three months using Kruskal–Wallis tests followed by Mann–Whitney U tests. These analyses have important limitations. Within-group significance does not estimate a treatment effect; an active-versus-control contrast does. Moreover, endpoint-only tests do not adjust for baseline imbalance, and Welch confidence intervals for mean differences are not inferentially aligned with rank-based tests.

For a definitive Q1 submission, the primary analysis should be repeated from participant-level data using a prespecified baseline-adjusted model. A linear model or suitable generalized model should compare groups at three months while adjusting for baseline attack frequency; a repeated-measures mixed-effects model can use all post-baseline observations and estimate group-by-time effects. Model assumptions and robust alternatives should be reported. Effect estimates with model-consistent 95% CIs should accompany p-values. Pairwise active-versus-sham contrasts and the exploratory active-versus-active contrast require multiplicity control. Secondary outcomes should be treated as supportive and adjusted for multiple testing. A responder analysis based on a prespecified reduction in migraine frequency should be added only if supported by the protocol and diary data.

Because participant-level data were not supplied for this rewrite, the results below retain the reported aggregate summaries and approximate, unadjusted endpoint CIs. They should be viewed as exploratory and replaced by the reanalysis described above. No imputation was needed for randomized participants because all reportedly provided follow-up data. Individuals excluded before randomization were not assigned or imputed to any trial group.

## **3. Results**

### **3.1 Participant flow and baseline characteristics**

Of 137 individuals assessed, 54 were excluded before randomization because they could not commit to the 24-session schedule. Eighty-three participants were randomized to supraorbital TENS (n = 28), infraorbital TENS (n = 27), or sham stimulation (n = 28). All reportedly received their assigned intervention, completed the schedule, and provided one- and three-month data. A completed CONSORT flow diagram must show numbers assessed, reasons for exclusion, intervention receipt, follow-up, and analysis by group.

The sample included 63 women (75.9%). Reported age and sex distributions were similar across groups (Table 1). Statistical tests of baseline balance are not informative in randomized trials and have therefore been removed. Clinically relevant baseline characteristics—migraine with/without aura, headache and migraine days, disease duration, acute medication use, and prior preventive-treatment exposure—were not available and should be added from source data.

Table 1. Reported baseline characteristics

| Characteristic                    | Supraorbital (n = 28) | Infraorbital (n = 27) | Sham (n = 28) |
|-----------------------------------|-----------------------|-----------------------|---------------|
| Age, years, mean (SD)             | 35.8 (7.6)            | 36.4 (6.9)            | 35.1 (7.2)    |
| Women/men, n                      | 21/7                  | 20/7                  | 22/6          |
| Migraine attacks/month, mean (SD) | 14.18 (3.95)          | 15.13 (4.51)          | 12.32 (4.99)  |

### 3.2 Primary outcome

Mean monthly migraine attack frequency decreased in both active groups and did not decrease in the sham group (Table 2). The unadjusted mean changes from baseline were 7.59 attacks/month in the supraorbital group, 8.97 in the infraorbital group, and +0.62 in the sham group. These change estimates are descriptive because SDs of within-participant change were unavailable.

At three months, the reported unadjusted mean differences versus sham were 6.35 attacks/month (approximate 95% CI, 8.46 to 4.24) for supraorbital TENS and 6.78 (8.86 to 4.70) for infraorbital TENS. The unadjusted supraorbital-versus-infraorbital difference was 0.43 (0.94 to 1.80). The original omnibus Kruskal-Wallis test was reported as  $p < 0.001$ . Baseline-adjusted model estimates and multiplicity-adjusted pairwise  $p$ -values require participant-level reanalysis.

Table 2. Monthly migraine attack frequency

| Group        | Baseline, mean (SD) | 1 month, mean (SD) | 3 months, mean (SD) | Mean change, baseline to 3 months |
|--------------|---------------------|--------------------|---------------------|-----------------------------------|
| Supraorbital | 14.18 (3.95)        | 7.68 (2.84)        | 6.59 (2.70)         | 7.59                              |
| Infraorbital | 15.13 (4.51)        | 9.19 (4.03)        | 6.16 (2.48)         | 8.97                              |
| Sham         | 12.32 (4.99)        | 14.27 (4.15)       | 12.94 (5.01)        | +0.62                             |

### 3.3 Secondary outcomes

Secondary outcomes showed lower three-month means in both active groups than in sham (Table 3). VAS decreased by 3.38 points in the supraorbital group, 3.17 points in the infraorbital group, and 0.57 points in the sham group. MIDAS and HIT-6 scores also decreased more in the active groups. Acute analgesic-use frequency and the reported MPQ score followed a similar pattern.

These findings are supportive rather than confirmatory. The estimates are unadjusted, multiple outcomes and pairwise contrasts were examined, and key measurement details are unavailable. In particular, MIDAS timing and MPQ scoring require verification before the corresponding results can be interpreted.

Table 3. Reported secondary outcomes

| Outcome                             | Group        | Baseline, mean (SD) | 1 month, mean (SD) | 3 months, mean (SD) | Mean change to 3 months |
|-------------------------------------|--------------|---------------------|--------------------|---------------------|-------------------------|
| VAS (0–10)                          | Supraorbital | 7.30 (1.29)         | 4.07 (0.76)        | 3.92 (0.97)         | 3.38                    |
|                                     | Infraorbital | 7.07 (1.47)         | 5.05 (1.24)        | 3.90 (1.33)         | 3.17                    |
|                                     | Sham         | 7.10 (1.11)         | 6.16 (1.44)        | 6.53 (1.02)         | 0.57                    |
| MIDAS                               | Supraorbital | 20.02 (5.22)        | 12.33 (3.94)       | 9.76 (3.12)         | 10.26                   |
|                                     | Infraorbital | 17.94 (6.30)        | 13.19 (4.62)       | 10.30 (2.07)        | 7.64                    |
|                                     | Sham         | 19.36 (4.05)        | 17.87 (5.20)       | 15.96 (5.39)        | 3.40                    |
| HIT-6                               | Supraorbital | 62.58 (4.78)        | 51.50 (3.61)       | 49.48 (4.84)        | 13.10                   |
|                                     | Infraorbital | 64.22 (5.93)        | 51.50 (5.39)       | 51.40 (3.01)        | 12.82                   |
|                                     | Sham         | 61.11 (3.74)        | 62.39 (4.20)       | 62.18 (4.13)        | +1.07                   |
| Acute analgesic-use occasions/month | Supraorbital | 14.62 (4.35)        | 6.21 (2.59)        | 5.02 (2.64)         | 9.60                    |
|                                     | Infraorbital | 14.72 (3.50)        | 8.57 (3.20)        | 5.65 (2.55)         | 9.07                    |
|                                     | Sham         | 14.36 (4.27)        | 12.72 (2.88)       | 12.06 (4.25)        | 2.30                    |
| Reported MPQ score                  | Supraorbital | 26.43 (6.49)        | 16.73 (5.74)       | 16.77 (5.08)        | 9.66                    |
|                                     | Infraorbital | 26.52 (6.23)        | 15.98 (4.63)       | 14.87 (5.34)        | 11.65                   |
|                                     | Sham         | 25.42 (6.40)        | 24.17 (6.49)       | 24.16 (4.23)        | 1.26                    |

### 3.4 Safety and tolerability

Itching was reported by 9/28 participants (32.1%) in the supraorbital group and 8/27 (29.6%) in the infraorbital group. Paresthesia or tingling was reported by 12/28 (42.9%) and 11/27 (40.7%), respectively. Events were described

as transient and did not lead to discontinuation. No serious adverse events were reported. Because safety ascertainment relied on open-ended inquiry, the incidence of mild or unexpected events may have been underestimated. Sham-group non-serious event counts should be reported explicitly.

## **4. Discussion**

### **4.1 Principal findings**

In this selected sample, both active TENS groups had lower mean monthly migraine attack frequency at three months than the inert sham group. Pain intensity, disability, headache impact, and acute medication-use outcomes showed a similar descriptive pattern. The active groups had comparable three-month means, but the study does not establish equivalent efficacy because it was not designed or powered for equivalence or non-inferiority.

The magnitude of the unadjusted differences is potentially clinically relevant. Nevertheless, effect estimates may be biased by baseline imbalance, an inert sham readily distinguishable from active stimulation, participant-reported outcomes, and selection of individuals able to attend 24 clinic sessions. Participant-level baseline-adjusted reanalysis is necessary before definitive efficacy claims are made.

### **4.2 Comparison with prior evidence**

The direction of the supraorbital findings is consistent with prior trials and reviews of external trigeminal nerve stimulation [5–10]. However, the effect sizes in the present report appear larger than those commonly observed in sham-controlled neuromodulation studies. This discrepancy may reflect differences in population, dose, endpoint definition, sham credibility, or analysis, and argues for caution rather than stronger causal inference.

Randomized evidence for infraorbital surface stimulation remains limited. The present findings justify further study of this target but do not isolate a site-specific physiological effect because the infraorbital intervention was compared with a supraorbital sham and participants could likely distinguish active from inactive stimulation. A future factorial or multi-arm design with sensory-matched sham at each site would address this problem more directly.

### **4.3 Outcome selection and analysis**

Current preventive-trial guidance favors monthly migraine days, moderate/severe headache days, or a prespecified responder rate [15]. The use of attack counts can be affected by the operational definition of when one attack ends and another begins and makes comparison with contemporary trials difficult. Future work should collect daily electronic diary data and report migraine days, headache days, attack frequency, acute medication days, and responder rates.

The original endpoint-only rank tests and within-group tests do not fully exploit the longitudinal randomized design. A baseline-adjusted model would provide a more appropriate treatment-effect estimate and improve precision. Model-consistent CIs, prespecified contrasts, and a transparent multiplicity strategy are essential for confirmatory interpretation.

### **4.4 Feasibility and clinical applicability**

Twenty-four supervised sessions over four weeks represent a substantial treatment burden. Thirty-nine percent of screened patients (54/137) declined or could not meet this schedule before randomization. Perfect attendance among randomized participants therefore reflects selection as well as feasibility among those enrolled. The results cannot be generalized to patients with limited access, work or caregiving constraints, or lower treatment motivation. Pragmatic trials should evaluate home-use protocols, fewer sessions, and remote adherence monitoring.

### **4.5 Strengths and limitations**

Strengths include a randomized three-arm design, a sham comparator, direct evaluation of two trigeminal targets, supervised treatment delivery, complete reported follow-up, and the use of several migraine-related outcomes.

The study also has substantial limitations:

1. The randomization report contains an unresolved internal inconsistency: fixed blocks of four are incompatible with balanced 1:1:1 allocation.
2. Ethics approval and prospective-registration identifiers were not supplied and must be verified before submission.
3. The available sample-size description was reconstructed after the study and does not constitute a valid prospective justification; the active-site comparison was underpowered.
4. The primary analysis was not adjusted for baseline and was not aligned with the longitudinal design.
5. Sham produced no sensation and was site-matched only to supraorbital stimulation, compromising masking and interpretation of the infraorbital contrast.
6. All efficacy outcomes were participant-reported and therefore vulnerable to expectation effects.
7. Thirty-nine percent of screened patients were excluded because of schedule burden, restricting external validity.

8. The primary endpoint was attack frequency rather than a guideline-preferred migraine-day endpoint.
9. VAS recall, MIDAS timing, and MPQ version/scoring were insufficiently documented.
10. Acute medication regimens were not standardized; medication days and changes in drug type or dose were not reported.
11. Adverse events were collected non-systematically.
12. Follow-up was limited to three months, the trial was single-center, and the sample was predominantly female.
13. Clinically important baseline variables and migraine-subtype results were not reported.

## 5. Conclusions

In aggregate, unadjusted outcomes at three months favored both supraorbital and infraorbital TENS over an inert supraorbital sham in participants selected for their ability to complete an intensive clinic schedule. These findings are promising but exploratory. They do not demonstrate equivalence between active sites and should not support strong clinical recommendations without verification of trial conduct and baseline-adjusted reanalysis of participant-level data. Larger, prospectively registered trials with credible site-matched sham stimulation, guideline-concordant diary outcomes, prespecified multiplicity control, and longer follow-up are warranted.

## Declarations

Ethics approval and consent to participate: [Insert verified committee name, approval identifier, approval date, and consent statement.]

Trial registration: [Insert registry, identifier, registration date, first enrollment date, and URL.]

Consent for publication: [Insert statement.]

Data availability: De-identified participant-level data, the protocol, statistical analysis plan, analysis code, and blank consent form should be made available in a suitable repository where permitted; otherwise, specify a justified controlled-access process.

Funding: [Insert verified funding statement.]

Competing interests: [Insert author-specific declarations, including any relationship with the device manufacturer.]

Author contributions: [Insert CRediT roles for each author.]

Acknowledgments: [Insert only contributors who meet acknowledgment criteria and have consented.]

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## Figure legends

Figure 1. CONSORT flow diagram. The final figure should report screening, exclusions with non-overlapping reasons, randomization, intervention receipt, follow-up, and analysis by group.

Figure 2. Mean monthly migraine attack frequency at baseline, one month, and three months by randomized group. Display individual observations or uncertainty intervals where possible; SD error bars alone do not represent uncertainty in group means.